ETHICAL HACKING-ITA1416

CO2_AT3 ERROR ANALYSIS TASK

SLOT-**B**

REGNO.**192512417**

Google Dorks Query Formulation

Mistake:

The student did not use proper Google search operators such as site:, filetype:, and quotation marks (" "), resulting in irrelevant search results.

Why improper search syntax leads to inaccurate footprinting:

- Returns unrelated websites and documents.
- Increases false positives.
- Wastes time analyzing irrelevant data.
- Reduces the accuracy of intelligence gathering.

Correct search queries:

- site:company.com
- site:company.com filetype:pdf
- "Company Name" site:company.com
- site:company.com filetype:xlsx
- intitle:"index of" site:company.com

These operators refine search results and help locate relevant public information.

2. Competitive Intelligence Source Validation

Mistake:

The student relied on blogs and unofficial forums without verifying the authenticity of the information.

Why unreliable data affects decision-making:

- Information may be outdated.
- Employee details may be incorrect.
- Organizational structure may be inaccurate.

- Leads to incorrect security assessments and poor business decisions.

Methods to verify information:

- Official company website
- LinkedIn company and employee profiles
- Government business registration databases
- Official press releases
- Trusted industry reports

These sources provide accurate and up-to-date information.

3. Passive vs Active Reconnaissance

Mistake:

The student performed active scanning before conducting passive footprinting.

Why passive reconnaissance is critical:

- Does not directly interact with the target.
- Reduces the chance of detection.
- Collects valuable public information safely.
- Helps plan targeted and efficient active scanning.

Correct sequence of footprinting steps:

1. Search engine reconnaissance
2. WHOIS lookup
3. DNS enumeration
4. Competitive intelligence gathering
5. Metadata analysis
6. Active scanning
7. Vulnerability assessment

4. WHOIS Analysis

Missing components:

- Domain expiry date

- Administrative contact
- Technical contact
- Name servers
- Registration and update dates

How these fields provide useful security insights:

- Expiry date identifies domains nearing expiration.
- Administrative contacts reveal responsible personnel.
- Technical contacts identify infrastructure managers.
- Name servers reveal DNS infrastructure.
- Registration dates indicate domain history.

How proper WHOIS analysis can reveal vulnerabilities:

- Identifies exposed administrative information.
- Detects weak DNS configurations.
- Reveals outdated domain registrations.
- Helps understand an organization's internet infrastructure.

5. Metadata Analysis

Mistake:

The student examined only the visible content of documents and ignored embedded metadata.

How metadata provides valuable intelligence:

Metadata may reveal:

- Author names
- Usernames
- Software versions
- Organization name
- Document creation and modification dates
- File paths
- System information

This information helps identify potential security weaknesses.

Correct method to extract metadata:

Use metadata analysis tools such as:

- ExifTool
- FOCA (Fingerprinting Organizations with Collected Archives)
- Exif Reader
- Microsoft Office Document Inspector
- PDF metadata viewers

Ethical use of metadata information:

- Identify exposed usernames and system details.
- Detect outdated software versions.
- Recommend removing sensitive metadata before publication.
- Improve document security and organizational awareness.
- Report findings responsibly to strengthen security without exploiting the information.